

Develop and Deploy a MeeGo Application



Open embedded platforms are becoming popular among end-users, and are also a key selling point for many vendors. New business models like application stores are thriving around these open platforms. Many industry biggies like Google and Samsung are jumping onto this bandwagon, to tap the growing market. The latest entrants—Nokia and Intel—have used MeeGo, a Linux-based platform. MeeGo is an entirely open source platform—from the kernel to the application framework, like UI libraries and tools. The MeeGo project is governed by the Linux Foundation (<http://www.linuxfoundation.org/>), and works like an open source project. In this article, we will take a quick look at this platform, and then develop a simple Qt application, which we can run on a MeeGo netbook.

The MeeGo project combines Nokia's Maemo and Intel's Moblin open source platforms to target devices like netbooks, hand-held devices, in-vehicle infotainment devices, etc. It comes with cross-platform development tools, and supports multiple hardware architectures like ARM, Intel's Atom, etc. MeeGo also boots faster, and has a small footprint, which is essential for embedded systems. MeeGo sports a new look and feel, with integrated social networking support.

Architecture

MeeGo has a three-layered architecture: operating system, middleware, and UX (user experience), as illustrated in Figure 1. The operating system layer provides the necessary hardware adaptation to run MeeGo. Based on the Linux kernel, it also contains the *Glibc* C and *libudev* libraries.

The middleware layer, based on the cross-platform Qt 4.x tool-kit, supports various usage models, and provides APIs to build applications. It also extends functionality through Web runtime APIs, which allows developers to build applications that can render

Web content like HTML, XML, SVG, etc. The UX layer has a specific 'look-and-feel' for netbooks, hand-held devices, and other products. It uses the whole screen to display controls, with no right-click contexts in touch-screen operation mode.

Though Maemo's UI was initially based on GTK+, it was later ported to Qt when Nokia acquired Trolltech, the creators of Qt. The Qt APIs are used from the C++ programming language.

Development environment

MeeGo provides a choice of environments to develop and debug Qt applications, both on Linux and Windows, as illustrated in Figure 2. These applications can then be packaged for installation on MeeGo devices.

A simple MeeGo application

Let's explore how to develop a Qt application on Linux, and install it on a MeeGo netbook. The page at http://wiki.meego.com/Hello_World_-_MeeGo_x86_development_on_Linux provides a step-by-step guide to developing applications for MeeGo. You can install the MeeGo SDK, which provides the necessary environment for MeeGo development, on your